III B.TECH – I SEMESTER

Construction technology and management

Open Elective-I | Code: 23A01505b | JNTUA R23 · B.Tech CSE

Course Code	Construction technology and management (OPEN ELECTIVE – I)	L	T	P	C
23A01505b		3	0	0	3
Course Objectives: **The objectives of this course are to make the student :** To understand project management fundamentals, organizational structures, and leadership principles in construction. To analyze manpower planning, equipment management, and cost estimation in civil engineering projects. To apply planning, scheduling, and project management techniques such as CPM and PERT. To evaluate various contract types, contract formation, and legal aspectsin construction management. To assess safety management practices, accident prevention strategies, and quality management systemsin construction.					
Course Outcomes (COs): **Upon successful completion of the course, students will be able to:** Understand (Cos)project management fundamentals, organizational structures, and leadership principles in construction. Analyze manpower planning, equipment management, and cost estimationin civil engineering projects. Apply planning, scheduling, and project management techniques such as CPM and PERT. Evaluate various contract types,					

contract formation, and legal aspectsin construction management. Assess safety management practices, accident prevention strategies, and quality management systems in construction.					
CO – PO Articulation Matrix Course Outcomes PO1 PO2 PO3 PO4 PO5 PO6 PO7 PO8 PO9 PO 10 PO 11 PO 12 PSO1 PSO2 CO -1 3 - - - - 2 - 2 2 - - - 3 3 **CO -2** - 3 - - 2 - - - - - - 2 3 3 **CO -3** - - 3 3 3 - - - - 2 - - 3 3 **CO -4** - - 3 3 3 - - 2 - - - - 3 3 **CO -5** - - - - - 3 3 3 2 - - - - 3					
UNIT – I					
Introduction: Project forms, Management Objectives and Functions; Organizational Chart of A Construction Company; Manager's Duties and Responsibilities; Public Relations; Leadership and Team - Work; Ethics, Morale, Delegation and Accountability.					
UNIT – II					
Man and Machine: Man-Power Planning, Training, Recruitment, Motivation, Welfare Measures and Safety Laws; Machinery for Civil Engineering., Earth Movers and Hauling Costs, Factors Affecting Purchase, Rent, and Lease of Equipment, and Cost Benefit Estimation.					
UNIT – III					
Planning, Scheduling andProject Management: Planning Stages, Construction Schedules and Project Specification, Monitoring and Evaluation; Bar-Chart, CPM, PERT, Network-formulation and Time Computation.					
UNIT – IV					
Contracts: Types of Contracts, formation of Contract – Contract					

Conditions – Contract forLabour, Material, Design, Construction – Drafting of Contract Documents Based On IBRD/ MORTH Standard Bidding Documents – Construction Contracts – Contract Problems – Arbitration and Legal Requirements Computer Applications in Construction Management: Software for Project Planning, Scheduling and Control.					
UNIT – V					
Safety Management – Implementation and Application of QMS in Safety Programs, ISO 9000 Series, Accident Theories, Cost of Accidents, Problem Areas in Construction Safety, Fall Protection, Incentives, Zero Accident Concepts, Planning for Safety, Occupational Health and Ergonomics.					
TEXT BOOKS:					
Construction Project Management, SK. Sears, GA. Sears, RH. Clough, John Wiley and Sons, 6th Edition, 2016. Construction Project Scheduling and Control by Saleh Mubarak, 4th Edition, 2019 Pandey, I.M (2021) Financial Management 12th edition. Pearson India Education Services Pvt. Ltd.					
REFRENCE BOOKS:					
Brien, J.O. andPlotnick, F.L., CPMin Construction Management, Mcgraw Hill, 2010. Punmia, B.C., andKhandelwal, K.K., Project Planning and control with PERT and CPM, Laxmi Publications, 2002. Construction Methods and Management: Pearson New International Edition 8 th Edition Stephens Nunnally. Rhoden, M and Cato B, Construction Management					

andOrganisational Behaviour, Wiley-Blackwell, 2016.					
Online Learning Resources:					
https://archive.nptel.ac.in/courses/105/104/105104161/ https://archive.nptel.ac.in/courses/105/103/105103093/					